

in software. Here, it is assumed that the user has characterised a thermistor and therefore has a lookup table of resistance vs temperature readily available. For the characteristics of thermistors, you may please refer to this month's EFY-CD.

The formula to measure the thermistor resistance is derived as follows:

Voltage across the thermistor

$$V_m = \frac{R1 \times V_{ref}}{R1 + R_t} \quad \text{----- (1)}$$

where $R1$ is the reference resistor (680-kilo-ohm), V_{ref} is the internal voltage reference (1.2V) and R_t is the thermistor resistance at a given temperature.

From Eq. (1), we can obtain thermistor resistance:

$$R_t = \frac{R1(V_{ref} - V_m)}{V_m} \quad \text{----- (2)}$$

Using the lookup table, consisting of characterised values of resistance vs temperature, along with a linear interpolation technique, you can get the equivalent temperature with a fairly high degree of precision.

Next, convert this value into a precisely scaled PWM waveform such that the average value of this waveform, in volts, is equal to the temperature measured in °C. As the PWM waveform swings between ground and V_{CC} , the average value is dependent on V_{CC} . In this case, since we are using a battery without regulation, V_{CC} changes over time as the battery discharges. Hence the program measures the battery voltage periodically and always uses this value to create the right PWM waveform. Thus a simple PWM is used to generate an accurate digital read-out on the multimeter over the entire lifetime of the battery.

Thermometer program

The `digit_thermo.c` thermometer program is written in 'C' language. It is given at the end of this article and also included in this month's EFY-CD.

Here is the outline of the program. After the initial variable declarations comes the `res_temp_lookup` table, containing the resistance vs temperature

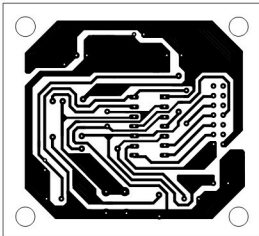


Fig. 14: Combined single-side, actual-size PCB layout for blinking LED and precision thermometer using MSP430 microcontroller

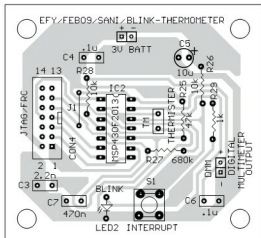


Fig. 15: Component layout for the PCB

values. About 99 values are present in this table to cover the temperature range from 0°C to 100°C accurately. The `getTemDegC` function is defined to fetch the correct value of temperature for a given thermistor resistance after interpolation between adjacent entries in the table.

The clock system is programmed to run both MCLK and SMCLK on the internal DCO at 2 MHz. ACLK runs on VLO. Ports P1.0, P1.1 and P1.3 are programmed to ensure proper ADC connectivity. ADC registers are programmed to use SMCLK.

The timer is set up such that it generates a 2kHz PWM waveform on port 1.2, which is fed to the digital multimeter. Switch S1 takes the system to sleep

mode or to active mode.

Variable `sysOn` is used as a flag to indicate the state of the controller system. This flag is toggled whenever an interrupt is generated using switch S1. The `sysOn` has a value '0' when S1 is pressed momentarily. When `sysOn` is '1', the system will be in active mode and keep measuring the temperature. When `sysOn` is '0', the system will be in sleep mode.

Even in active mode, the system normally enters low-power mode 1 (or LPM1 mode), where MCLK and the CPU are 'off' but SMCLK and ACLK are running. SMCLK is needed to keep the PWM output active to enable continuous reading of the temperature.

The watchdog timer is programmed to wake-up the system approximately every second. On wake-up, the system measures V_{CC} value and the thermistor resistance through the ADC channels. The resistance is converted into temperature. These values are used to set up an appropriate duty cycle on the PWM waveform whose average value equals temperature in °C.

Note that this system consumes about 500 µA of current when measuring and displaying temperature. This means that it will work for more than 300 hours continuously with a CR2032 (3V lithium cell). In sleep mode, the system consumes just about 0.1 µA. Hence the battery will last longer for intermittent temperature measurements. Additionally, the low sleep-mode current eliminates the need for a separate power switch as the system can be put to sleep mode without worrying about draining the battery.

Construction

Combined actual-size, single-side PCB layout for the blinking LED and thermometer is shown in Fig. 14 and its